

# Brutal Series B12 - Physical & Chemical Changes / Exponents & Powers / Transportation / Symmetry

Each question has 6 to 8 options; exactly ONE is correct. Answers are scattered - there is NO pattern. Every question needs several steps - read carefully and work it out fully.

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1. An iron gate gains 0.45 g of mass each week because rust forms on it. Rusting is which kind of change, and by what mass does the gate increase in 8 weeks? [\[Physical & Chemical Changes\]](#)
  - (A) Chemical change; 36 g
  - (B) Physical change; mass stays the same
  - (C) Physical change; 3.6 g
  - (D) No change; 0 g
  - (E) Chemical change; 0.45 g
  - (F) Chemical change; 4.05 g
  - (G) Chemical change; mass increases by 3.6 g
  - (H) Chemical change; 2.7 g
  
2. A spreadsheet cell computes  $(2^5 \times 2^3)$  divided by  $2^2$ . Write the result as a single power of 2 and give its value. [\[Exponents & Powers\]](#)
  - (A)  $2^8 = 256$
  - (B)  $2^6 = 64$
  - (C)  $2^4 = 16$
  - (D)  $2^{10} = 1024$
  - (E)  $2^5 = 64$
  - (F)  $2^6 = 32$
  - (G)  $2^7 = 128$
  
3. A resting heart beats 72 times each minute and pumps 70 mL of blood per beat. How many litres does it pump in one hour? [\[Transportation in Animals & Plants\]](#)
  - (A) 252 L
  - (B) 30.24 L
  - (C) 30240 L
  - (D) 5.04 L
  - (E) 302.4 L
  - (F) 302.4 mL
  - (G) 3024 L
  
4. A regular polygon has rotational symmetry of order 9. Through what smallest angle (in degrees) can it be turned to look the same, and how many lines of symmetry does it have? [\[Symmetry\]](#)
  - (A) 40 deg; 9 lines
  - (B) 40 deg; 18 lines
  - (C) 9 deg; 9 lines
  - (D) 45 deg; 9 lines
  - (E) 36 deg; 9 lines
  - (F) 40 deg; 8 lines

5. Burning a candle has two parts: wax melting (near the flame) and wax burning (in the flame). Which statement is fully correct? [\[Physical & Chemical Changes\]](#)

- (A) Melting is physical, burning is also physical
- (B) Melting is chemical, burning is physical
- (C) Neither is a change
- (D) Melting is chemical, burning is chemical
- (E) Both are physical changes
- (F) Melting is physical, burning is chemical

6. Simplify  $(3^4 \times 3^2)$  divided by  $(3^3 \times 3)$  and give the final number. [\[Exponents & Powers\]](#)

- (A) 3
- (B) 18
- (C) 12
- (D) 6
- (E) 1
- (F) 81
- (G) 9
- (H) 27

7. Blood has about 5 million red blood cells in each cubic millimetre. A drop of blood is 0.05 cubic millimetre. About how many red cells are in the drop? [\[Transportation in Animals & Plants\]](#)

- (A) 2500000
- (B) 100000
- (C) 2500
- (D) 25000
- (E) 5000000
- (F) 50000
- (G) 250000
- (H) 250000000

8. An equilateral triangle is rotated about its centre. How many times in one full turn (including the start position back again) does it look exactly the same? [\[Symmetry\]](#)

- (A) 3 times
- (B) 9 times
- (C) 6 times
- (D) 4 times
- (E) 0 times
- (F) 2 times

9. In a school lab 5 changes are shown: ice melting, milk turning to curd, paper being torn, a biscuit being baked, salt dissolving in water. How many are chemical changes? [\[Physical & Chemical Changes\]](#)

- (A) 0
- (B) Cannot be decided
- (C) 2
- (D) 4
- (E) 1
- (F) 5
- (G) 3

**10.** The mass of one dust grain is about  $5 \times 10^{-6}$  g. Write the mass of 4000 such grains in standard form.

[Exponents & Powers]

- (A)  $2 \times 10^{-3}$  g
- (B)  $2 \times 10^{-1}$  g
- (C)  $8 \times 10^{-3}$  g
- (D)  $5 \times 10^{-3}$  g
- (E)  $20 \times 10^{-3}$  g
- (F)  $2 \times 10^{-2}$  g

**11.** Water rises in the xylem of a plant. In one experiment water climbs 24 cm in 8 minutes at a steady rate. How far does it climb in 1 hour 20 minutes? [Transportation in Animals & Plants]

- (A) 2.4 m and that equals 24 cm
- (B) 48 cm
- (C) 24 cm
- (D) 240 cm
- (E) 3 cm
- (F) 180 cm

**12.** A shape returns to its first look after every 72-degree turn. What is its order of rotational symmetry?

[Symmetry]

- (A) 72
- (B) 3
- (C) 5
- (D) 8
- (E) 10
- (F) 6
- (G) 2

**13.** A blacksmith heats an iron strip until it glows red, then bends it into a hook and lets it cool. Counting only what happened, which is true? [Physical & Chemical Changes]

- (A) Both physical and chemical changes occurred
- (B) The iron became a new metal
- (C) Only a physical change occurred
- (D) Cooling reversed a chemical change
- (E) Only a chemical change occurred
- (F) No change occurred

**14.** Find the value of  $5^0 + 2^3 - 3^2 + 4^1$  and choose the correct total. [Exponents & Powers]

- (A) 4
- (B) 0
- (C) 5
- (D) 13
- (E) 8
- (F) 6

**15.** An adult body holds about 5 L of blood, and the heart circulates all of it in about 1 minute. How many times is the whole blood volume circulated in 1 hour? [\[Transportation in Animals & Plants\]](#)

- (A) 30 times
- (B) 60 times
- (C) 12 times
- (D) 5 times
- (E) 600 times
- (F) 300 times

**16.** The capital letters H, I, O, X, S are checked for line symmetry. How many of these five letters have AT LEAST one line of symmetry? [\[Symmetry\]](#)

- (A) 3
- (B) 5
- (C) 2
- (D) 1
- (E) H and O only
- (F) All except S and X
- (G) 4
- (H) 0

**17.** Photosynthesis takes in carbon dioxide and water and makes glucose and oxygen. A plant makes 18 g of glucose and releases 16 g of oxygen one day. What type of change is this, and what total mass of products formed? [\[Physical & Chemical Changes\]](#)

- (A) Physical change; 34 g
- (B) No change; 0 g
- (C) Chemical change; 34 g of products
- (D) Chemical change; 2 g
- (E) Chemical change; 18 g
- (F) Chemical change; 16 g
- (G) Physical change; 2 g

**18.** A sheet of paper is folded in half 7 times. Each fold doubles the number of layers. How many layers are there after 7 folds? [\[Exponents & Powers\]](#)

- (A) 64
- (B) 14
- (C) 49
- (D) 49
- (E) 32
- (F) 128

**19.** A patient is given fluid through a drip at 30 drops per minute, and 20 drops make 1 mL. How many millilitres are given in 2 hours? [\[Transportation in Animals & Plants\]](#)

- (A) 36 mL
- (B) 180 mL
- (C) 18 mL
- (D) 60 mL
- (E) 3600 mL
- (F) 1800 mL
- (G) 90 mL
- (H) 360 mL

**20.** A regular hexagon has rotational symmetry of order 6 and the same number of lines of symmetry. What is the sum of its number of lines of symmetry and its smallest angle of rotation (in degrees)? [\[Symmetry\]](#)

- (A) 12
- (B) 96
- (C) 72
- (D) 60
- (E) 6
- (F) 66
- (G) 126
- (H) 36

**21.** A galvanised bucket is iron coated with zinc. The zinc coating weighs 24 g and protects the iron from rusting. If  $\frac{1}{4}$  of the zinc wears off, how much zinc is left, and why is galvanising done? [\[Physical & Chemical Changes\]](#)

- (A) 18 g left; to colour the bucket
- (B) 6 g left; to make the bucket heavier
- (C) 18 g left; the zinc layer stops the iron from rusting
- (D) 18 g left; to start rusting faster
- (E) 12 g left; to stop rusting
- (F) 18 g left; to make iron magnetic
- (G) 20 g left; to stop rusting
- (H) 6 g left; to stop rusting

**22.** Express 0.00045 in standard form, then state its exponent of 10. [\[Exponents & Powers\]](#)

- (A)  $45 \times 10^{-5}$ , exponent -5
- (B)  $4.5 \times 10^4$ , exponent 4
- (C)  $0.45 \times 10^{-3}$ , exponent -3
- (D)  $4.5 \times 10^{-3}$ , exponent -3
- (E)  $4.5 \times 10^{-4}$ , exponent 4
- (F)  $4.5 \times 10^{-2}$ , exponent -2
- (G)  $4.5 \times 10^{-5}$ , exponent -5
- (H)  $4.5 \times 10^{-4}$ , exponent -4

- 23.** Plants lose water as vapour through tiny pores called stomata. A leaf loses 2 mL of water per hour by this process. What is this process called, and how much is lost in a 12-hour day? [\[Transportation in Animals & Plants\]](#)
- (A) Respiration; 24 mL
  - (B) Transpiration; 2 mL
  - (C) Transpiration; 6 mL
  - (D) Transpiration; 14 mL
  - (E) Transpiration; 24 mL
  - (F) Photosynthesis; 24 mL
  - (G) Transpiration; 240 mL
  - (H) Translocation; 24 mL
- 24.** A square has 4 lines of symmetry and rotational order 4. A rectangle (not a square) has how many lines of symmetry and what rotational order? [\[Symmetry\]](#)
- (A) 0 lines; order 2
  - (B) 4 lines; order 4
  - (C) 2 lines; order 2
  - (D) 2 lines; order 1
  - (E) 1 line; order 2
  - (F) 2 lines; order 4
  - (G) 3 lines; order 2
- 25.** When dilute acid is poured on baking soda, bubbles of gas rush out and the mixture feels cold. Which TWO signs of a chemical change are seen here? [\[Physical & Chemical Changes\]](#)
- (A) A gas is produced and there is a change in temperature
  - (B) The mixture freezes solid
  - (C) A colour change and a smell
  - (D) Light is given out and a gas forms
  - (E) No signs of a chemical change
  - (F) Only the temperature changes
- 26.** Compare the two numbers  $3^5$  and  $5^3$ . State which is larger and by how much. [\[Exponents & Powers\]](#)
- (A)  $3^5$  is larger by 18
  - (B) They are equal
  - (C)  $3^5$  is larger by 368
  - (D)  $5^3$  is larger by 18
  - (E)  $3^5$  is larger by 118
  - (F)  $5^3$  is larger by 118
  - (G)  $5^3$  is larger by 2
- 27.** The wall of the left ventricle is thicker than the right because it pumps blood all around the body. If the left wall is 12 mm and the right is 4 mm, the left is how many times as thick? [\[Transportation in Animals & Plants\]](#)
- (A) 6 times
  - (B) 0.33 times
  - (C) 1.5 times
  - (D) 8 times
  - (E) 12 times
  - (F) 4 times
  - (G) 2 times
  - (H) 3 times

**28.** A wheel design looks the same after a turn of 30 degrees. What is its order of rotational symmetry, and how many equal spokes could create this? [\[Symmetry\]](#)

- (A) Order 12; 12 spokes
- (B) Order 12; 6 spokes
- (C) Order 30; 30 spokes
- (D) Order 360; 360 spokes
- (E) Order 10; 10 spokes
- (F) Order 6; 6 spokes
- (G) Order 12; 30 spokes

**29.** A jeweller melts 200 g of gold, pours it into a ring mould, and lets it set. The next day it is still 200 g of gold. What does this tell us about the change? [\[Physical & Chemical Changes\]](#)

- (A) It is physical because mass increased
- (B) It is chemical because heat was used
- (C) Mass should have decreased
- (D) It is chemical because the shape changed
- (E) Melting always makes a new substance
- (F) It is chemical because the gold melted
- (G) It is physical because no new substance formed and mass stayed the same

**30.** Simplify  $(a^5 \times a^3)$  divided by  $a^2$  for a non-zero number  $a$ , and write the result as a single power.

[\[Exponents & Powers\]](#)

- (A)  $a^8$
- (B)  $a^{16}$
- (C)  $a^6$
- (D)  $a^{10}$
- (E)  $a^2$
- (F)  $a^4$
- (G)  $a^{30}$

**31.** Arteries carry blood away from the heart and veins carry it back. A nurse lists 5 vessels: 2 carry blood away from the heart, the rest carry it back. What fraction of the listed vessels are veins? [\[Transportation in Animals & Plants\]](#)

[\[Plants\]](#)

- (A)  $\frac{4}{5}$
- (B)  $\frac{2}{3}$
- (C)  $\frac{1}{5}$
- (D)  $\frac{1}{2}$
- (E)  $\frac{2}{5}$
- (F)  $\frac{3}{5}$

**32.** A regular polygon has an interior angle of 140 degrees. Each exterior angle equals the smallest angle of rotational symmetry. How many sides and lines of symmetry does it have? [\[Symmetry\]](#)

- (A) 7 sides; 7 lines
- (B) 9 sides; 18 lines
- (C) 8 sides; 8 lines
- (D) 9 sides; 8 lines
- (E) 6 sides; 6 lines
- (F) 10 sides; 10 lines
- (G) 9 sides; 9 lines

**33.** Curdling, ripening of fruit, and digestion of food are listed. A student says all three make new substances. Is the student right, and what does that make these changes? [\[Physical & Chemical Changes\]](#)

- (A) Yes; they are reversible physical changes
- (B) Yes; but only ripening is chemical
- (C) No; all three are physical
- (D) Yes; all three are chemical changes
- (E) No; only digestion is chemical
- (F) Yes; but they are physical
- (G) No; none make new substances
- (H) No; only curdling is chemical

**34.** A colony of bacteria triples every hour. If it starts with 2 bacteria, how many are there after 4 hours? Write as a product with a power and the value. [\[Exponents & Powers\]](#)

- (A)  $3^4 = 81$
- (B)  $6^4 = 1296$
- (C)  $2 \times 4^3 = 128$
- (D)  $2 \times 3^4 = 96$
- (E)  $2 \times 3^4 = 81$
- (F)  $2 \times 3^3 = 54$
- (G)  $2 \times 3^4 = 162$
- (H)  $2 + 3^4 = 83$

**35.** Counting a pulse gives the heartbeat. A child's pulse is counted as 18 beats in 15 seconds. What is the heart rate in beats per minute? [\[Transportation in Animals & Plants\]](#)

- (A) 60 beats per minute
- (B) 72 beats per minute
- (C) 270 beats per minute
- (D) 108 beats per minute
- (E) 18 beats per minute
- (F) 90 beats per minute
- (G) 36 beats per minute
- (H) 48 beats per minute

**36.** The digits 0, 1, 8 are checked. Which statement about their rotational symmetry of order 2 (half-turn) is correct? [\[Symmetry\]](#)

- (A) Only 1 and 8 look the same
- (B) Only 8 looks the same
- (C) All three change after a half-turn
- (D) 0, 1 and 8 all look the same after a half-turn
- (E) Only 0 looks the same
- (F) None look the same after a half-turn



- 37.** Crystals of copper sulphate are grown by letting a blue solution slowly evaporate, leaving solid crystals. Is crystallisation physical or chemical, and can the crystals be re-dissolved? [\[Physical & Chemical Changes\]](#)
- (A) Physical; they become a new salt
  - (B) Physical; yes, they can be re-dissolved
  - (C) Chemical; no
  - (D) Physical; only if heated again
  - (E) Chemical; they become a new salt
  - (F) Neither; the water is destroyed
- 38.** Evaluate  $(-2)^4 + (-2)^3$  and give the result. [\[Exponents & Powers\]](#)
- (A) -16
  - (B) 24
  - (C) -8
  - (D) 8
  - (E) 0
  - (F) 16
  - (G) 32
  - (H) -24
- 39.** Phloem carries food made in the leaves to the rest of the plant. If a plant makes 40 g of sugar in its leaves and 35% is sent to the roots, how many grams reach the roots? [\[Transportation in Animals & Plants\]](#)
- (A) 35 g
  - (B) 140 g
  - (C) 1.4 g
  - (D) 16 g
  - (E) 14 g
  - (F) 26 g
  - (G) 3.5 g
  - (H) 4 g
- 40.** A figure has rotational symmetry of order 8. List the four smallest turn angles (in degrees) at which it matches itself. [\[Symmetry\]](#)
- (A) 45, 90, 135, 180
  - (B) 45, 90, 180, 360
  - (C) 45, 135, 225, 315
  - (D) 45, 90, 135, 225
  - (E) 90, 180, 270, 360
  - (F) 40, 80, 120, 160
- 41.** Setting of cement, formation of clouds, and digestion of bread are given. Identify the ONE change in this list that is physical. [\[Physical & Chemical Changes\]](#)
- (A) All three are physical
  - (B) Digestion of bread
  - (C) Formation of clouds
  - (D) None are physical
  - (E) Setting of cement and digestion
  - (F) Setting of cement

**42.** The distance to a star is  $3 \times 10^{13}$  km and a probe travels  $6 \times 10^4$  km in a day. Roughly how many days, in standard form, to reach the star? [\[Exponents & Powers\]](#)

- (A)  $0.5 \times 10^9$  days
- (B)  $5 \times 10^8$  days
- (C)  $2 \times 10^8$  days
- (D)  $5 \times 10^9$  days
- (E)  $5 \times 10^7$  days
- (F)  $5 \times 10^{17}$  days
- (G)  $18 \times 10^8$  days

**43.** Blood is roughly 55% plasma and 45% cells by volume. In a 4000 mL blood volume, how many millilitres more plasma than cells are there? [\[Transportation in Animals & Plants\]](#)

- (A) 40 mL
- (B) 4000 mL
- (C) 200 mL
- (D) 400 mL
- (E) 2200 mL
- (F) 1800 mL
- (G) 100 mL

**44.** A kite has exactly 1 line of symmetry. An isosceles triangle also has exactly 1. If a pattern is made of 7 kites and 5 isosceles triangles, what is the total number of lines of symmetry of all these shapes together?

[\[Symmetry\]](#)

- (A) 12
- (B) 35
- (C) 7
- (D) 24
- (E) 13
- (F) 5
- (G) 2

**45.** A 50 g piece of magnesium ribbon burns completely in air and the white ash left weighs 83 g. Why did the mass go UP, and what change is this? [\[Physical & Chemical Changes\]](#)

- (A) Chemical change; water was added
- (B) Mass cannot increase in burning
- (C) Chemical change; oxygen from the air joined the magnesium
- (D) Physical change; oxygen joined the metal
- (E) Physical change; the ribbon expanded
- (F) Chemical change; some magnesium escaped

**46.** Simplify  $7^9$  divided by  $7^9$  and also  $7^9$  divided by  $7^7$ . Give both answers. [\[Exponents & Powers\]](#)

- (A) First = 0; second = 7
- (B) First = 1; second = 343
- (C) First = 1; second = 49
- (D) First =  $7^0$ ; second =  $7^7$
- (E) First = 7; second = 49
- (F) First = 1; second = 14
- (G) First = 0; second = 49

47. Root hairs absorb water from the soil. If 1 cm of root has 200 root hairs and a root is 35 cm long, how many root hairs are there, written in standard form? [\[Transportation in Animals & Plants\]](#)

- (A)  $7 \times 10^3$
- (B)  $70 \times 10^2$
- (C)  $35 \times 10^2$
- (D)  $7 \times 10^5$
- (E)  $7 \times 10^4$
- (F)  $7 \times 10^2$

48. A regular polygon's smallest angle of rotational symmetry is 24 degrees. How many sides does it have, and is 24 a factor of 360? [\[Symmetry\]](#)

- (A) 15 sides; 24 is not whole
- (B) 10 sides; yes
- (C) 16 sides; yes
- (D) 12 sides; yes
- (E) 15 sides; no, 24 is not a factor of 360
- (F) 15 sides; no sides exist
- (G) 15 sides; yes, 24 is a factor of 360

49. Four kitchen events: water boiling to steam, bread going mouldy, sugar dissolving in tea, an egg being fried. How many are reversible physical changes? [\[Physical & Chemical Changes\]](#)

- (A) 3
- (B) 0
- (C) Only the egg
- (D) Cannot be decided
- (E) All four
- (F) 1
- (G) 4
- (H) 2

50. Find the value of  $\{(5^2)^2\}$  divided by  $5^3$  and write it as a power of 5 with its value. [\[Exponents & Powers\]](#)

- (A)  $5^4 = 625$
- (B)  $5^3 = 125$
- (C)  $5^2 = 25$
- (D)  $5^7 = 78125$
- (E)  $5^5 = 3125$
- (F)  $5^0 = 1$
- (G)  $5^1 = 5$

51. The heart of a person beats about 72 times a minute. About how many times does it beat in one full day (24 hours)? [\[Transportation in Animals & Plants\]](#)

- (A) 103680
- (B) 72000
- (C) 103680 hours
- (D) 10368
- (E) 259200
- (F) 4320
- (G) 1036800
- (H) 1728

**52.** When you fold a shape along a line of symmetry, the two halves match exactly. A regular octagon is folded along all its lines of symmetry one at a time. How many such folds are possible? [\[Symmetry\]](#)

- (A) 4
- (B) 8
- (C) 6
- (D) 16
- (E) 10
- (F) 12
- (G) 1

**53.** A nail left in damp air rusts; a dry nail kept in oil does not. After 30 days the damp nail's mass rises by 9%. If it started at 20 g, what is its new mass, and why didn't the oiled nail rust? [\[Physical & Chemical Changes\]](#)

- (A) 20.9 g; oil keeps air away
- (B) 18.2 g; oil keeps air away
- (C) 21.8 g; oil keeps air and moisture away from the iron
- (D) 22.0 g; oil keeps water in
- (E) 1.8 g; oil keeps air away
- (F) 21.8 g; oil makes iron heavier

**54.** Write 64 as a power of 2, as a power of 4, and as a power of 8. Which single statement is fully correct?

[\[Exponents & Powers\]](#)

- (A)  $64 = 2^7 = 4^3 = 8^2$
- (B)  $64 = 2^6 = 4^3 = 8^2$
- (C)  $64 = 2^5 = 4^3 = 8^2$
- (D)  $64 = 2^6 = 4^4 = 8^2$
- (E)  $64 = 2^6 = 4^3 = 8^3$
- (F)  $64 = 2^8 = 4^4 = 8^2$
- (G)  $64 = 2^6 = 4^2 = 8^2$

**55.** Two plants are watered. Plant A pulls up water at 5 mL per hour, Plant B at 8 mL per hour. After a 9-hour day, how much more water has Plant B pulled up than Plant A? [\[Transportation in Animals & Plants\]](#)

- (A) 13 mL
- (B) 45 mL
- (C) 117 mL
- (D) 24 mL
- (E) 27 L
- (F) 3 mL
- (G) 72 mL
- (H) 27 mL

- 56.** A windmill has 4 blades placed with rotational symmetry of order 4 but NO line symmetry. What is the angle between successive blades, and how many lines of symmetry does it have? [\[Symmetry\]](#)
- (A) 120 deg apart; 0 lines
  - (B) 45 deg apart; 0 lines
  - (C) 60 deg apart; 0 lines
  - (D) 90 deg apart; 2 lines
  - (E) 90 deg apart; 1 line
  - (F) 360 deg apart; 0 lines
  - (G) 90 deg apart; 4 lines
  - (H) 90 deg apart; 0 lines
- 57.** Dissolving washing soda in water lets it be recovered by evaporation, but burning a matchstick cannot be undone. This contrast best shows the difference between: [\[Physical & Chemical Changes\]](#)
- (A) Two reversible changes
  - (B) Two chemical changes
  - (C) Heating and cooling
  - (D) Two physical changes
  - (E) Melting and freezing
  - (F) A reversible chemical and an irreversible physical change
  - (G) A physical (reversible) change and a chemical (irreversible) change
- 58.** A number is  $2^3 \times 3^2 \times 5$ . A second number is  $2 \times 3^2 \times 5^2$ . Which is larger and by how much? [\[Exponents & Powers\]](#)
- (A) The second is larger by 45
  - (B) The second is larger by 9
  - (C) The second is larger by 450
  - (D) They are equal
  - (E) The first is larger by 9
  - (F) The second is larger by 90
  - (G) The first is larger by 360
- 59.** Blood pressure is often written as two numbers like 120/80. A healthy reading is 120 over 80. By what percentage is the top number greater than the bottom number? [\[Transportation in Animals & Plants\]](#)
- (A) 25%
  - (B) 20%
  - (C) 30%
  - (D) 33%
  - (E) 40%
  - (F) 50%
  - (G) 60%
  - (H) 150%

**60.** A regular polygon has 20 lines of symmetry. What is its smallest angle of rotational symmetry, and what is the sum of all its interior angles? [\[Symmetry\]](#)

- (A) 20 deg; 3240 deg
- (B) 18 deg; 3060 deg
- (C) 9 deg; 3240 deg
- (D) 18 deg; 360 deg
- (E) 18 deg; 2880 deg
- (F) 18 deg; 3240 deg
- (G) 36 deg; 3240 deg

**61.** Souring of milk takes about 6 hours at room temperature but only 1.5 hours in warm weather. The souring itself is which kind of change, and how many times FASTER is it in warm weather? [\[Physical & Chemical Changes\]](#)

- (A) Chemical change; 1.5 times faster
- (B) Physical change; the same speed
- (C) Chemical change; 2 times faster
- (D) Physical change; 4 times faster
- (E) Chemical change; 4.5 times faster
- (F) Chemical change; 6 times faster
- (G) Physical change; 2 times faster
- (H) Chemical change; 4 times faster

**62.** Simplify  $(10^6 \times 10^{-2})$  divided by  $10^3$  and give the value as an ordinary number. [\[Exponents & Powers\]](#)

- (A) 0.1
- (B) 10000
- (C) 0.01
- (D) 10
- (E) 1000000
- (F) 1000
- (G) 100
- (H) 1

**63.** Lymph is a pale fluid that drains from tissues and joins the blood. If the body forms 3 L of lymph in a day and 2.4 L returns to the blood, what fraction does NOT immediately return? [\[Transportation in Animals & Plants\]](#)

- (A)  $\frac{2}{5}$
- (B)  $\frac{3}{5}$
- (C)  $\frac{1}{5}$
- (D)  $\frac{2}{3}$
- (E)  $\frac{4}{5}$
- (F)  $\frac{1}{4}$

**64.** A shape made of two overlapping squares (an 8-point star) looks the same after a 45-degree turn. What is its order of rotational symmetry? [\[Symmetry\]](#)

- (A) 8
- (B) 5
- (C) 2
- (D) 45
- (E) 6
- (F) 16
- (G) 45/8

**65.** Five processes: melting butter, electrolysis-free rusting of iron, sublimation of camphor, fermentation of dough, sharpening a pencil. Which set lists ONLY the physical changes? [\[Physical & Chemical Changes\]](#)

- (A) Sublimation of camphor, fermentation of dough
- (B) Rusting of iron, fermentation of dough
- (C) Rusting, sublimation, sharpening
- (D) Melting butter, sublimation of camphor, sharpening a pencil
- (E) All five
- (F) Melting butter, rusting of iron
- (G) Melting butter, fermentation, sharpening
- (H) Fermentation, sublimation, sharpening

**66.** If  $2^x = 32$ , find  $x$ , then compute  $3^x$ . [\[Exponents & Powers\]](#)

- (A)  $x = 5$ , so  $3^x = 81$
- (B)  $x = 5$ , so  $3^x = 243$
- (C)  $x = 4$ , so  $3^x = 81$
- (D)  $x = 5$ , so  $3^x = 125$
- (E)  $x = 5$ , so  $3^x = 15$
- (F)  $x = 6$ , so  $3^x = 729$
- (G)  $x = 5$ , so  $3^x = 243$  but  $x = 4$

**67.** A doctor records that a heart pumps 5 L of blood per minute. The lungs receive all of it. How many litres pass through the lungs in 24 minutes? [\[Transportation in Animals & Plants\]](#)

- (A) 24 L
- (B) 29 L
- (C) 120 L
- (D) 240 L
- (E) 600 L
- (F) 100 L

**68.** Among the shapes: scalene triangle, isosceles triangle, equilateral triangle, which has the MOST lines of symmetry, and how many? [\[Symmetry\]](#)

- (A) Equilateral triangle; 6 lines
- (B) Scalene triangle; 1 line
- (C) Equilateral triangle; 1 line
- (D) Isosceles triangle; 2 lines
- (E) Equilateral triangle; 3 lines
- (F) Isosceles triangle; 1 line

**69.** A signboard reads: 'CO<sub>2</sub> turns clear limewater milky.' In a test, gas from a burning candle turns limewater milky. What does this prove the candle made, and is burning physical or chemical? [\[Physical & Chemical Changes\]](#)

- (A) Carbon dioxide; chemical
- (B) Nitrogen; chemical
- (C) Carbon dioxide; neither
- (D) Water only; physical
- (E) Oxygen; physical
- (F) Oxygen; chemical

**70.** The thickness of a coin is  $2 \times 10^{-3}$  m. A stack of 1500 coins is how tall, in standard form (metres)? [\[Exponents & Powers\]](#)

- (A)  $3 \times 10^0$  m
- (B)  $3 \times 10^3$  m
- (C)  $30 \times 10^{-1}$  m
- (D)  $3 \times 10^2$  m
- (E)  $3 \times 10^{-1}$  m
- (F)  $1.5 \times 10^0$  m
- (G)  $3 \times 10^1$  m
- (H)  $3 \times 10^{-3}$  m

**71.** Valves in the heart and veins stop blood flowing backwards. A vein has valves every 4 cm along its 60 cm length. How many gaps of 4 cm are there, and so how many valves at the ends and between? [\[Transportation in Animals & Plants\]](#)

- (A) 15 gaps, so 30 valves
- (B) 15 gaps, so 14 valves
- (C) 15 gaps, so 16 valves
- (D) 16 gaps, so 16 valves
- (E) 240 gaps, so 241 valves
- (F) 15 gaps, so 15 valves
- (G) 14 gaps, so 15 valves
- (H) 60 gaps, so 61 valves

**72.** A clock face is marked at every hour. Treating the 12 marks as a symmetric pattern, what is the angle between two neighbouring hour marks, and the order of rotational symmetry? [\[Symmetry\]](#)

- (A) 15 deg; order 12
- (B) 30 deg; order 6
- (C) 60 deg; order 6
- (D) 30 deg; order 12
- (E) 30 deg; order 24
- (F) 30 deg; order 12 but angle 12
- (G) 36 deg; order 12
- (H) 12 deg; order 30



- 73.** A tray of water is left in the freezer and becomes ice; later it is left out and melts back to water. Across the whole freeze-then-melt cycle, the substance: [\[Physical & Chemical Changes\]](#)
- (A) Is chemical when freezing, physical when melting
  - (B) Is destroyed and remade
  - (C) Loses mass when melted
  - (D) Becomes a new substance when melted
  - (E) Stays as water the whole time, so both steps are physical
  - (F) Gains mass when frozen
  - (G) Is chemical both times
- 74.** Simplify the expression  $(2^4 \times 3^4)$  and write it as a single power. [\[Exponents & Powers\]](#)
- (A)  $6^4$
  - (B)  $6^2$
  - (C)  $36^4$
  - (D)  $6^8$
  - (E)  $6^4$  but value 24
  - (F)  $2^4 + 3^4$
  - (G)  $12^4$
- 75.** Tall trees move water from roots to the topmost leaves. If sap rises 1.5 m every hour and the tree is 18 m tall, how many hours for water to reach the top? [\[Transportation in Animals & Plants\]](#)
- (A) 1.5 hours
  - (B) 24 hours
  - (C) 9 hours
  - (D) 6 hours
  - (E) 12 hours
  - (F) 27 hours
  - (G) 18 hours
- 76.** A regular pentagon and a regular decagon (10 sides) are compared. How many more lines of symmetry does the decagon have, and what is the decagon's smallest rotation angle? [\[Symmetry\]](#)
- (A) 5 more lines; 10 deg
  - (B) 5 more lines; 36 deg
  - (C) 2 more lines; 36 deg
  - (D) 5 fewer lines; 36 deg
  - (E) 5 more lines; 60 deg
  - (F) 10 more lines; 36 deg
  - (G) 5 more lines; 72 deg
- 77.** When iron reacts with copper sulphate solution, the blue colour fades and a reddish-brown coating of copper appears on the iron. Which clues here point to a chemical change? [\[Physical & Chemical Changes\]](#)
- (A) No clues; it is physical
  - (B) Only the colour change
  - (C) The iron melting
  - (D) A gas is released
  - (E) The colour change and the new copper coating
  - (F) The blue colour staying the same
  - (G) The temperature drop only
  - (H) Only the new coating

**78.** Evaluate  $4^3$  divided by  $2^4$ , writing the top with base 2 first. [\[Exponents & Powers\]](#)

- (A) 1
- (B) 0.5
- (C) 32
- (D) 64
- (E) 4
- (F) 16
- (G) 2
- (H) 8

**79.** Capillaries are the narrowest blood vessels. If a capillary is 8 micrometres wide and a red blood cell is also about 8 micrometres, and  $1\text{ mm} = 1000\text{ micrometres}$ , how many such capillaries laid side by side span 1 mm? [\[Transportation in Animals & Plants\]](#)

- (A) 80
- (B) 625
- (C) 12.5
- (D) 1250
- (E) 125
- (F) 8000
- (G) 100

**80.** A pattern repeats every  $\frac{1}{6}$  of a full turn. Express the smallest rotation angle as a fraction of 360 and as degrees, and state the order. [\[Symmetry\]](#)

- (A)  $\frac{1}{6}$  of 360 = 6 deg; order 6
- (B)  $\frac{1}{6}$  of 360 = 60 deg; order 12
- (C)  $\frac{1}{6}$  of 360 = 60 deg; order 3
- (D)  $\frac{1}{6}$  of 360 = 36 deg; order 10
- (E)  $\frac{1}{6}$  of 360 = 60 deg; order 6
- (F)  $\frac{1}{6}$  of 360 = 60 deg; order 60

**81.** A student lists signs of chemical change and ticks: gas released, light/heat given out, new colour, change in shape. How many of the four ticks are CORRECT signs of a chemical change? [\[Physical & Chemical Changes\]](#)

- (A) All except gas
- (B) Only colour change
- (C) 3
- (D) 2
- (E) 1
- (F) 0
- (G) 4

**82.** A virus count starts at  $10^2$  and multiplies by 10 each day. Write the count after 5 days in standard form. [\[Exponents & Powers\]](#)

- (A)  $1 \times 10^{52}$
- (B)  $2 \times 10^7$
- (C)  $1 \times 10^7$
- (D)  $1 \times 10^7$  but base 5
- (E)  $1 \times 10^6$
- (F)  $5 \times 10^2$
- (G)  $1 \times 10^{10}$
- (H)  $1 \times 10^5$

**83.** Plants need both xylem and phloem. A student says xylem carries water UP and phloem carries food in MANY directions. In a diagram of 6 tubes, 4 are xylem. What percent are phloem? [\[Transportation in Animals & Plants\]](#)

- (A) About 40%
- (B) About 60%
- (C) About 66%
- (D) About 33%
- (E) About 50%
- (F) About 67%

**84.** A shape has rotational symmetry of order 6. Through how many degrees, in total, must it be turned to pass through ALL its matching positions and return to the very start? [\[Symmetry\]](#)

- (A) 120 deg
- (B) 216 deg
- (C) 180 deg
- (D) 720 deg
- (E) 300 deg
- (F) 6 deg
- (G) 60 deg
- (H) 360 deg

**85.** A 12 g strip of magnesium is only half burned; the burned half becomes oxide while the unburned half is still metal. The burned half gained 4 g of oxygen. What is the total mass now? [\[Physical & Chemical Changes\]](#)

- (A) 10 g
- (B) 24 g
- (C) 4 g
- (D) 14 g
- (E) 16 g
- (F) 8 g

**86.** Find the value of  $(3^3 + 3^2)$  divided by  $3^2$  as a fraction in lowest terms. [\[Exponents & Powers\]](#)

- (A) 1
- (B) 12
- (C) 6
- (D) 3
- (E)  $10/3$
- (F) 4
- (G)  $9/3$

**87.** The pulse felt at the wrist matches the heartbeat. If a runner's pulse rises from 70 to 126 beats per minute, by what percentage did it increase? [\[Transportation in Animals & Plants\]](#)

- (A) 44%
- (B) 8%
- (C) 80%
- (D) 126%
- (E) 180%
- (F) 56%
- (G) 70%

**88.** The letters of the word 'MOM' are written. Considering the whole word's symmetry, which line of symmetry does 'MOM' have? [\[Symmetry\]](#)

- (A) Two diagonal lines
- (B) No line of symmetry
- (C) A vertical and a diagonal line
- (D) A vertical line down the middle
- (E) Rotational only, no line
- (F) Both a vertical and a horizontal line
- (G) Only a diagonal line

**89.** Tearing paper, then burning the torn pieces, happens in that order. Classify the two steps in order.

[\[Physical & Chemical Changes\]](#)

- (A) Physical, then chemical
- (B) Both reversible
- (C) Physical, then physical
- (D) Neither is a change
- (E) Physical, then reversible physical
- (F) Chemical, then reversible physical
- (G) Chemical, then physical
- (H) Chemical, then chemical

**90.** The Sun is about  $1.5 \times 10^8$  km away and Mars is about  $2.28 \times 10^8$  km from the Sun. How much farther from the Sun is Mars, in standard form (km)? [\[Exponents & Powers\]](#)

- (A)  $78 \times 10^6$  km
- (B)  $0.78 \times 10^8$  km
- (C)  $7.8 \times 10^6$  km
- (D)  $7.8 \times 10^8$  km
- (E)  $3.78 \times 10^8$  km
- (F)  $7.8 \times 10^7$  km

**91.** Deoxygenated blood is dark red; oxygenated blood is bright red. In the heart, the right side handles deoxygenated blood. If a model shows 240 mL on the right and the right holds 40% of the model's blood, how much blood is in the whole model? [\[Transportation in Animals & Plants\]](#)

- (A) 600 mL
- (B) 6000 mL
- (C) 144 mL
- (D) 400 mL
- (E) 288 mL
- (F) 240 mL

- 92.** A regular polygon is such that its order of rotational symmetry equals its number of sides, which is 15. Its smallest rotation angle times its number of lines of symmetry equals what? [\[Symmetry\]](#)
- (A) 36
  - (B) 24
  - (C) 15
  - (D) 360
  - (E) 180
  - (F) 3600
- 93.** A rusted iron sheet weighs 56 g; scraping off all the rust leaves 50 g of clean iron and 6 g of rust powder. The 6 g of rust is heavier than the iron it replaced because: [\[Physical & Chemical Changes\]](#)
- (A) Oxygen from the air added to the iron to form the rust
  - (B) Rust is the same as iron
  - (C) Rust contains no iron
  - (D) Water has no mass
  - (E) Iron is lighter than air
  - (F) The rust soaked up extra iron
- 94.** Simplify  $\{(2^3)^2 \times 2\}$  divided by  $2^4$  to a single power of 2 and give its value. [\[Exponents & Powers\]](#)
- (A)  $2^2 = 4$
  - (B)  $2^3 = 9$
  - (C)  $2^3 = 6$
  - (D)  $2^5 = 32$
  - (E)  $2^3 = 8$
  - (F)  $2^1 = 2$
  - (G)  $2^6 = 64$
  - (H)  $2^4 = 16$
- 95.** Water absorbed by roots is carried up and most is lost by transpiration. A plant absorbs 500 mL of water in a day and loses 97% by transpiration. How much water is kept by the plant? [\[Transportation in Animals & Plants\]](#)
- (A) 485 mL
  - (B) 30 mL
  - (C) 150 mL
  - (D) 50 mL
  - (E) 15 mL
  - (F) 485 percent
- 96.** Two shapes are described: Shape P has 0 lines but rotational order 3; Shape Q has 1 line but rotational order 1. Which has GREATER rotational symmetry, and what is the difference in their orders? [\[Symmetry\]](#)
- (A) Shape Q; difference of 1
  - (B) Shape P; difference of 4
  - (C) Shape P; difference of 3
  - (D) Shape P; difference of 1
  - (E) Shape P; difference of 2
  - (F) They are equal; difference 0
  - (G) Shape Q; difference of 3
  - (H) Shape Q; difference of 2

**97.** Cutting a cake into 8 equal slices and baking the cake batter are compared. Which is physical, which is chemical, and how many degrees of one slice's central angle is cut? [\[Physical & Chemical Changes\]](#)

- (A) Cutting is physical, baking is chemical; each slice is 45 deg
- (B) Cutting is chemical, baking is physical; 45 deg
- (C) Both chemical; 45 deg
- (D) Cutting is physical, baking is chemical; 8 deg
- (E) Cutting is chemical, baking is chemical; 45 deg
- (F) Cutting is physical, baking is chemical; 60 deg

**98.** Evaluate  $2^5$  minus  $5^2$  and give the difference. [\[Exponents & Powers\]](#)

- (A) 0
- (B) 7
- (C) 3
- (D) 10
- (E) -7
- (F) 25
- (G) 57

**99.** A pump-like heart moves 4500 mL of blood every minute. How many litres of blood does it move in 20 minutes? [\[Transportation in Animals & Plants\]](#)

- (A) 90 L
- (B) 9 L
- (C) 90 mL
- (D) 90000 L
- (E) 45 L
- (F) 9000 L

**100.** A general parallelogram that is not a rectangle or a rhombus has how many lines of symmetry and what order of rotational symmetry? [\[Symmetry\]](#)

- (A) 2 lines; order 1
- (B) 1 line; order 2
- (C) 0 lines; order 4
- (D) 0 lines; order 1
- (E) 1 line; order 1
- (F) 4 lines; order 4
- (G) 0 lines; order 2